

EL OSCILADOR ARMÓNICO VESIÓN ALGEBRAICA

$$H = \frac{P^2}{2m} + \frac{m\omega^2}{2} R^2$$

Definición

$$A \equiv aP + bR$$

$$B \equiv A^\dagger \equiv a^*P + b^*R \quad (a, b \in \mathbb{C})$$

$$AB = |a|^2 P^2 + ab^* PR + ba^* RP + |b|^2 R^2$$

$$BA = |a|^2 P^2 + a^* b PR + b^* a RP + |b|^2 R^2$$

$$\text{Elijamos: } a = \frac{1}{\sqrt{2m}}, \quad b = -i\sqrt{\frac{m}{2}}\omega$$

$$\Rightarrow \begin{cases} AB = H + \frac{\hbar\omega}{2} \\ BA = H - \frac{\hbar\omega}{2} \end{cases}$$

$$[A, B] = \hbar\omega$$

Si ϕ es vector propio unitario de $N \equiv BA$
 $N\phi = \lambda\phi$

Muestre que $\lambda \geq 0$

$$\Rightarrow BA\phi = \lambda\phi$$

$$(\phi, BA\phi) = \lambda(\phi, \phi)$$

$$(A\phi, A\phi) = \lambda\|\phi\|^2$$

$$\lambda = \|A\phi\|^2 \geq 0$$

$$\begin{aligned} ab^* &= i\frac{\omega}{2} \\ ba^* &= -i\frac{\omega}{2} \end{aligned} \quad \left. \vphantom{\begin{aligned} ab^* &= i\frac{\omega}{2} \\ ba^* &= -i\frac{\omega}{2} \end{aligned}} \right\} \begin{aligned} ab^* PR + ba^* RP &= \\ i\frac{\omega}{2} [PR - RP] &= i\frac{\omega}{2} \frac{\hbar}{i} \\ &= \frac{\hbar\omega}{2} \end{aligned}$$

- $[N, A] = -\hbar \omega A$
 $[BA, A] = B \overset{\circ}{[A, A]} + [B, A] A$
 $= -\hbar \omega A$

- $[N, B] = \hbar \omega B$

- $[N, A^\nu] = -\nu \hbar \omega A^\nu$

$\Rightarrow$

$$\begin{aligned} [N, A^{\nu+1}] &= [N, A^\nu] A + A^\nu [N, A] \\ &= -\nu \hbar \omega A^{\nu+1} - \hbar \omega A^{\nu+1} \\ &= -(\nu+1) \hbar \omega A^{\nu+1} \end{aligned}$$

- $[N, B^\nu] = \nu \hbar \omega B^\nu \quad \leftarrow$

- $NA^\nu = A^\nu (N - \nu \hbar \omega)$

- $NB^\nu = B^\nu (N + \nu \hbar \omega)$

Sea: $Q_\nu = B^\nu A^\nu$

Dq: $Q_\nu = Q_{\nu-1} [N - (\nu-1) \hbar \omega]$

$\Rightarrow$ de $NA^\nu = A^\nu (N - \nu \hbar \omega)$

$$NA^{\nu-1} = A^{\nu-1} (N - (\nu-1) \hbar \omega)$$

$$BAA^{\nu-1} = A^{\nu-1} (N - (\nu-1) \hbar \omega)$$

Multiplicando por $B^{\nu-1}$:

$$B^\nu A^\nu = B^{\nu-1} A^{\nu-1} (N - (\nu-1) \hbar \omega)$$

- $Q_\nu \phi = (\lambda - (\nu-1) \hbar \omega) Q_{\nu-1} \phi$

$$(\widehat{B^\nu A^\nu} \phi) = (\lambda - (\nu-1) \hbar \omega) (\widehat{B^{\nu-1} A^{\nu-1}} \phi)$$

$\Rightarrow$ $\|A^\nu \phi\|^2 = (\lambda - (\nu-1) \hbar \omega) \|A^{\nu-1} \phi\|^2$

- $H = \frac{P^2}{2m} + m\omega^2 R^2$

$$AB = H + \frac{\hbar\omega}{2} \quad \widetilde{BA}^N = H - \frac{\hbar\omega}{2}$$

$$\Rightarrow \boxed{H = N + \frac{\hbar\omega}{2}}$$

- $\|A^\nu \phi\|^2 = (\lambda - (\nu - 1)\hbar\omega) \|A^{\nu-1} \phi\|^2$

- Mq: $A^\nu B^\nu = A^{\nu-1} B^{\nu-1} (N + \nu\hbar\omega)$

$$NB^\nu = B^\nu (N + \nu\hbar\omega)$$

$$BAB^\nu = B^\nu (N + \nu\hbar\omega)$$

$$AB^\nu = B^{\nu-1} (N + \nu\hbar\omega)$$

$$\boxed{A^\nu B^\nu = A^{\nu-1} B^{\nu-1} (N + \nu\hbar\omega)}$$

- Mostrar que ϕ es valor propio de $B^\nu A^\nu$ como de $A^\nu B^\nu$.

$$[A, B] = \hbar\omega$$

$$AB - BA = \hbar\omega$$

$$AB\phi = BA\phi + \hbar\omega\phi$$

$$AB\phi = N\phi + \hbar\omega\phi$$

$$= (\lambda + \hbar\omega)\phi$$

- Verificar:

$$A^{\nu-1}\phi \neq 0 \rightarrow \lambda \geq (\nu - 1)\hbar\omega$$

$$\text{De (*): } \|A^\nu \phi\|^2 \geq 0$$

$$\rightarrow \lambda \geq (\nu - 1)\hbar\omega$$

$$\|A^\nu \phi\|^2 = (\lambda - (\nu - 1)\hbar\omega) \|A^{\nu-1} \phi\|^2$$

- Verificar:

$$\lambda < (\nu - 1)\hbar\omega \rightarrow A^{\nu-1}\phi = 0$$

$$\rightarrow \phi = 0$$

$$\text{De (*): } \Rightarrow \|A^{\nu-1}\phi\| = 0$$

$$\|A^{v-2}\phi\| = 0$$

$$\vdots$$

$$\|A^{v-v}\phi\| = 0$$

$$\rightarrow \phi = 0$$

$$\|A^v\phi\|^2 = (\lambda - (v-1)\hbar\omega) \|A^{v-1}\phi\|^2$$

$$\|A^{v-1}\phi\|^2 = (\lambda - (v-2)\hbar\omega) \|A^{v-2}\phi\|^2$$

$$\vdots$$

- Verifique que dado el número real λ , existe un número entero v_0 tal que:

$$(v_0 - 1)\hbar\omega < \lambda \leq v_0\hbar\omega$$

λ : valor propio de ϕ

$$\begin{matrix} v_0 - 1 < \frac{\lambda}{\hbar\omega} \leq v_0 \\ \in \mathbb{Z} & \in \mathbb{R} & \in \mathbb{Z} \end{matrix}$$

(Un número real existe entre dos números enteros)

$$\circ \text{---} \bullet \quad v\hbar\omega$$

$$\text{---} \circ \text{---} \bullet \text{---} \text{---} \quad (v-2)\hbar\omega$$

$$\text{---} \circ \text{---} \bullet \text{---} \text{---} \quad (v-1)\hbar\omega \quad (v+1)\hbar\omega$$

- Mostrar que: $A^{v_0}\phi \neq 0$

$$\|A^{v_0}\phi\|^2 = [\lambda - (v_0 - 1)\hbar\omega] \|A^{v_0-1}\phi\|^2$$

Si:

$$A^{v_0}\phi = 0 \quad \wedge \quad \lambda > (v_0 - 1)\hbar\omega$$

$$\Rightarrow \phi = 0$$

(lo cual es absurdo)

$$\therefore A^{v_0}\phi \neq 0$$

- Verifique que:

i) Si $\lambda < v_0\hbar\omega \rightarrow A^{v_0}\phi = 0$

$$\|A^{\nu_0+1}\phi\|^2 = \underbrace{(\lambda - \nu_0 \hbar \omega)}_{<0} \|A^{\nu_0}\phi\|^2$$

$$\Rightarrow A^{\nu_0}\phi = 0$$

ii) $\lambda = \nu_0 \hbar \omega$

$$N\phi = \lambda\phi = \nu_0 \hbar \omega \phi$$

Operador $\nwarrow$ entero
número

iii) $A^{\nu_0+1}\phi = 0$

Como $\lambda = \nu_0 \hbar \omega \rightarrow A^{\nu_0+1}\phi = 0$

Sea $\phi_0 = \frac{A^{\nu_0}\phi}{\|A^{\nu_0}\phi\|}$

Verificar que: $N\phi_0 = 0$

$$\frac{BA^{\nu_0}\phi}{\|A^{\nu_0}\phi\|} = \frac{BA^{\nu_0+1}\phi}{\|A^{\nu_0}\phi\|} = 0$$

- Verificar que: $NA^m\phi_0 = -m\hbar\omega A^m\phi$

$\uparrow$
Vectores propios

$$NA^\nu = A^\nu (N - \nu \hbar \omega)$$

$$NA^\nu\phi_0 = A^\nu (N - \nu \hbar \omega)\phi_0$$

$$\begin{aligned} NA^m\phi_0 &= A^m N\phi_0 - m\hbar\omega A^m\phi_0 \\ &= -m\hbar\omega A^m\phi_0 \end{aligned}$$

- Verificar que: $NB^m\phi_0 = m\hbar\omega B^m\phi_0$

- Sea $\phi_n \equiv \frac{B^n\phi_0}{\|B^n\phi_0\|}$

Verificar que: $N\phi_n = n\hbar\omega\phi_n$

$$N \left(\frac{B^m \phi_0}{\|B^m \phi_0\|} \right)_{\phi_n} = m\hbar\omega \left(\frac{B^m \phi_0}{\|B^m \phi_0\|} \right)_{\phi_n}$$

N: operador número

- $$NA\phi_n = (n-1)\hbar\omega A\phi_n$$

$$NA = A(N - \hbar\omega)$$

$$\begin{aligned} NA\phi_n &= A(N - \hbar\omega)\phi_n \\ &= AN\phi_n - A\hbar\omega\phi_n \\ &= \hbar\omega(n-1)A\phi_n \end{aligned}$$

A: operador de bajada (aniquilación)

B: operador de subida (creación)

- $$NB\phi_n = (n+1)\hbar\omega B\phi_n$$

- $$\|B^n \phi_0\|^2 = n\hbar\omega \|B^{n-1} \phi_0\|^2$$

- Mostrar que: $\|B^n \phi_0\|^2 = n!(\hbar\omega) \|\phi_0\|^2$

$$\|B^n \phi_0\|^2 = n\hbar\omega(n-1)\hbar\omega \dots$$

$$\dots (n - (n-1))\hbar\omega \|B^{n-n} \phi_0\|^2 = n!(\hbar\omega)^n \|\phi_0\|^2$$

Verificar que:
$$A\phi_n = (n\hbar\omega)^{1/2} \phi_{n-1}$$

$$B\phi_n = [(n+1)\hbar\omega]^{1/2} \phi_{n+1}$$

$$NB^m \phi_0 = m\hbar\omega B^m \phi_0$$

$$BAB^m \phi_0 = m \hbar \omega B^m \phi_0$$

$$AB^m \phi_0 = m \hbar \omega B^{m-1} \phi_0$$

$$\frac{AB^m \phi_0}{\|B^m \phi_0\|} = \frac{m \hbar \omega B^{m-1} \phi_0}{(m \hbar \omega)^{1/2} \|B^{m-1} \phi_0\|}$$

$$A\phi_m = \sqrt{m \hbar \omega} \phi_{m-1}$$

$$\begin{array}{l} \vdots \\ \text{_____} B^2 \phi_n \\ \text{_____} B \phi_n \\ \text{_____} \phi_n \\ \text{_____} A \phi_n \\ \text{_____} A^2 \phi_n \\ \vdots \\ \text{_____} A^n \phi_n = \phi_0 \end{array}$$

- Vq. las soluciones de $H\Phi = E\Phi$ son los ϕ_n .

$$H\phi_n = \left(n + \frac{1}{2}\right) \hbar \omega \phi_n$$

$$H = N + \frac{\hbar \omega}{2} \Rightarrow H\phi_n = N\phi_n + \frac{\hbar \omega}{2} = \left(n + \frac{1}{2}\right) \hbar \omega$$